Fiscal Year:	FY 2019	**Task Last Updated:**	FY 10/16/2018
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Project Title:	NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	01/01/2016	**End Date:**	12/31/2020
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	NOTE: Continuation of "NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL," grant number NNX10AU19G. Funding is for Dr. Kelton's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted. Main project: Thermophysical and structural properties and solidification behavior of supercooled glass forming liquids as a member of THERMOLAB; the Science Coordinator is Prof. Dr. Hans Fecht, Ulm University, Institute of Micro and Nanomaterials, Albert-Einstein-Allee 47, Ulm Germany. Similar measurements on undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL); the Science Coordinator is Dr. Dirk Holland-Moritz, Deutsches Zentrum für Luft- und Raumfahrt (DLR), Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany. Subprojects: 1. Ground-Based Experimental Program: Synchrotron X-ray studies of liquid structures are made at the Advanced Photon Source using the ESL (electrostatic levitation) chamber at Washington University (WU-BESL), which has been optimized for scattering studies. These are correlated with thermophysical property data obtained on ground with WU-BESL and using the Material Science Laboratory Electromagnetic Levitator (MSL-EML) on the International Space Station (ISS). Terrestrial maximum undercooling and induction time to recalescence data are obtained in ESL, where stirring ensures that nucleation is not diffusion controlled. This data will allow an evaluation of diffusion in the upcoming MSL-EML ISS undercooling studies. 2. Thermophysical Property Measurements: ISS. Specific heat, density, surface tension and viscosity measurements will be made of supercooled glass-forming liquids and of Ti-Zr-Ni liquids with different Ni concentrations as a function of the temperature using the MSL-EML facility. These will be correlated with x-ray synchrotron diffraction studies of the supercooled liquids and ground-based measurements made using WU-BESL The purpose is to determine the existence and nature of liquid/liquid phase transitions that may be associated with chemical and topological ordering and a possible connection with liquid fragility as well as to obtain indicators in the specific heat of a cooperative cross over behavior in the shear viscosity at high temperatures. 3. Effect of diffusion on nucleation: The importance of long-range diffusion on nucleation will be evaluated using the MSL-EML facility located on the ISS. The critical items of interest are the maximum undercooling and the induction time to recalescence in isothermal holds at temperatures below the liquidus temperature. Together with the results from the thermophysical property measurements and an MHD convection model results this will allow us to analyze how the maximum undercooling is influenced by fluid flow. These results of the micro-g experiments will be compared with the results of ground-based ESL measurements and with predictions from Kelton’s coupled-flux model for nucleation. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: 1. Ground Based Experiment Program. Measurements of the density, surface tension and viscosity and synchrotron x-ray diffraction measurements of the liquid structure in the supercooled state using WU-BESL. Ground-based measurements of the maximum undercooling and induction time to recalescence in isothermal holds will also be made using WU-BESL. These will be used to evaluate the nature of ordering on phase transitions in the supercooled liquid and fragility. 2. Long Duration Microgravity Experiments. Measurements of thermophysical properties, specific heat (modulated measurements), surface tension, viscosity and density, and also electrical resistivity will be made as a function of temperature using the MSL-EML on the ISS. These will be coupled with x-ray and elastic neutron diffraction measurements of the liquid structure factor (WP 2) and ground based studies of the density and viscosity to study the role of ordering on novel liquid/liquid phase transitions and to determine the nature of these transitions. Maximum undercooling and time to recalesce measurements as a function of temperature and stirring will be compared with terrestrial ESL measurements to assess the importance of diffusion on nucleation, allowing Kelton's coupled flux model for nucleation to be evaluated experimentally.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Product advancement in industries such as information technology, aerospace, automotive, and basic and new materials manufacturing requires advanced materials development. In the past, this was often carried out through a laborious process of guided trial and error. The vast increases in computing power have now opened the door for realistic modeling approaches, which have the potential to greatly improve the efficiency of materials development. In the US, this has become a central focus of research in materials science, encapsulated in the Materials Genome Initiative. However, progress is frequently hampered by the lack of relevant thermophysical property data needed for the modeling. Further, key processes such as crystal nucleation from the liquid or other phase transitions within the liquid, are often poorly understood, making it difficult to realistically model them. In many cases, the experimental data and a deeper understanding of some processes cannot be obtained from terrestrial experiments due to interactions with sample containers at the temperatures of interest. Even containerless based methods, such as electrostatic levitation, cannot be used to accurately investigate these. The ISS experiments that are being carried out under the support of this grant are allowing these data to be obtained.

Task Progress:	Summary of activities during the past year: 1. Analysis of the data from ISS experiments In 2017 a series of experiments were carried out on three alloy liquids, Ti39.5Zr39.5Ni21, Zr64Ni36, and Cu50Zr50, under the microgravity environment of the electromagnetic levitation facility (EML-MSL) facility on board the International Space Station (ISS). Analyzing and giving a theoretical interpretation of the experimental data obtained for crystal nucleation and thermophysical properties of the liquid have been the focus of our research group over the past year. The nucleation studies on the ISS were complemented by extensive ground-based experiments on Ti39.5Zr39.5Ni21 and Zr64Ni36; studies of Cu50Zr50 are now in progress. One important result is that the nucleation parameters obtained by analyzing the data within the framework of the classical theory of nucleation indicate that the icosahedral phase nucleates much more easily in the Ti39.5Zr39.5Ni21 liquid than does the crystalline phase from the Zr64Ni36 liquid. This supports our earlier conclusions from ground-based structural studies that the dominant short-range order (SRO) in Ti39.5Zr39.5Ni21 is more similar to the icosahedral phase and the SRO in Zr64Ni36 liquid is less similar to its crystal phase. This is a robust conclusion, confirming that the structure of the liquid plays a critical role in the formation of the crystal phases. Since this is not considered in any currently used model for nucleation, our results demonstrate the need for a new theoretical framework. A second result is that for both alloy liquids crystal nucleation occurs more readily in the microgravity environment of the ISS occurs than it does on the Earth. Given that both sets of experiments were performed under high vacuum (approaches 10(-8) torr) this result is unlikely to be the cause of sample contamination during processing and may be due to the different levels of stirring between the two experimental studies. However, it may also be an artifact of the limited statistics allowed in the ISS experiments. The surface tension, the viscosity, the specific heat, and the electrical resistivity were measured for Ti39.5Zr39.5Ni21 and Zr64Ni36 liquids under microgravity and terrestrial experiments. The surface tension and viscosity under these two conditions were in general agreement, although some discrepancies were observed in a few cases. However, these differences were reconciled when factors such as the spinning of the sample and the vacuum conditions were taken into account. The electrical resistivity could only be measured on the ISS. Due to difficulties in making such measurements in metallic liquids at high temperatures, these are the first data obtained for these systems. The data were well explained by the extended Ziman theory, especially the negative temperature coefficients (decrease in resistivity with increasing temperature). The temperature dependence of the resistivity changed abruptly at a temperature that coincides with a change in the viscosity. This crossover temperature, T_A, correlates with the onset of cooperativity in the liquid dynamics. It is very exciting that a dynamical property could be correlated with the electrical resistivity for the first-time and could be explained theoretically. Molecular dynamics simulations for the partial structure factors of the liquid made by this group played an important role in analyzing these data. The analysis of modulation calorimetry data in the ISS experiments to obtain the specific heat and the emissivity of the supercooled liquids is in progress. 2. Other ISS-related activities a) Experimental studies of a metallic glass-forming liquid (Vit106) were completed in September and October 2018 during Batch 1.3 on the ISS. More extensive modulation calorimetry and nucleation data were obtained under a high vacuum environment. The analysis of this data has started. b) Preparations for Batch 3 experiments are in progress. My research team (QUASI) is the Principal Investigator (PI) for two alloys in this batch (Cu47Zr47Al6 and Zr80Pt20) and is a co-PI with the ICOPROSOL research team for studies of Ti45Zr45Ni10. The ability to process these samples in the EML was demonstrated in earlier parabolic flight experiments. Sample coupling and emissivity measurements are planned for these samples by MUSC around the end of this year. To facilitate those studies, five samples for each alloy composition were prepared by QUASI according to specifications and delivered to MUSC in September 2018. 3. Other related activities a) Ground-based structural studies of liquids have remained an integral part of the ISS experiments for a long time. A couple of controversies have been brewing in the scientific literature for about a decade regarding liquid structures. These are (i) whether reliable estimates of the volume expansion coefficients of liquids can be made from structural data and (ii) whether the structure of liquids follow a “fractal” behavior. Our earlier studies showed that the answers to both questions are negative. Despite that, a number of publications (especially under high pressure) have appeared in high-profile journals (e.g., PNAS, PRL) supporting a positive view. We have recently addressed this question from extensive structural studies of a large number (of order 30) metallic liquids and have demonstrated that neither of the conclusions from the other groups is correct. They are artifacts of an incorrect analysis of the experimental data. The dominance of certain partial structure factors in the scattering experiments mask the behavior of the bulk liquid and are responsible for the incorrect conclusions. b) A structural signature of the dynamical crossover has been elusive thus far. However, it was recently demonstrated by my group that upon cooling the liquid, the growth of a low q (i.e., low angle) feature in the second peak of the X-ray structure factor, S(q) accelerates at a temperature near T_A. This indicates an increasing growth of medium range order in the liquid.
Bibliography Type:	Description: (Last Updated: 05/20/2025)
Articles in Peer-reviewed Journals	Gangopadhyay AK, Kelton KF. "A re-evaluation of thermal expansion measurements of metallic liquids and glasses from x-ray scattering experiments." J Chem Phys. 2018 May 28;148(20):204509. https://doi.org/10.1063/1.5032319 ; PubMed PMID: 29865799 , May-2018
Articles in Peer-reviewed Journals	Dai R, Ashcraft R, Kelton KF. "A possible structural signature of the onset of cooperativity in metallic liquids." J Chem Phys. 2018 May 28;148(20):204502. https://doi.org/10.1063/1.5026801 ; PubMed PMID: 29865850 , May-2018
Articles in Peer-reviewed Journals	Johnson ML, Gibbons PC, Vogt AJ, Kelton KF. "Metastable phase selection from undercooled Zr77Rh23 liquid alloys." Journal of Alloys and Compounds. 2017 Nov 25;725:1217-22. https://doi.org/10.1016/j.jallcom.2017.07.241 , Nov-2017